GABRIELLA'S MATH LAB · DECIMALS WORKBOOK

SECTION 6 · SUPPLEMENTAL

Multiplying Decimals

Use equal groups and place value to multiply, then learn the shortcut for placing the decimal point.

SKILL
One new idea at a time.

SHOW HOW
Watch a problem solved step by step.

TRY ONE
Finish a problem with help.

YOUR TURN
Solve problems on your own.

6.1	Multiplying by 10, 100, and 1,000	☐ done	2
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Name _____ Started _____

Supplemental: Georgia's Grade 5 decimal standard (5.NR.4) covers adding and subtracting decimals. Multiplying decimals is a Grade 6 standard (6.NR.1); Skill 6.1 connects to 5.NR.1.2 (powers of 10).

6.1 Multiplying by 10, 100, and 1,000

Goal: I can multiply by 10, 100, or 1,000 by moving digits to bigger places.

SHOW HOW

Multiply 3.45 × 100.

Ones 1		Tenths 0.1	Hundredths 0.01
3	.	4	5

→

Hundreds 100	Tens 10	Ones 1		Tenths 0.1
3	4	5	.	

1. 100 = 10 × 10, so each digit becomes worth **10 times, then 10 times again**: it moves **2 places to the left**.
2. 3 ones → 3 hundreds, 4 tenths → 4 tens, 5 hundredths → 5 ones.
3. **3.45 × 100 = 345**. Shortcut: count the zeros (100 has 2), move the decimal point that many places to the **right**.

Why it works: each place is 10 times the place to its right (Section 1). × 10 moves digits 1 place, × 100 moves 2, × 1,000 moves 3. Fill empty places with zeros: 0.7 × 100 = 70.

LET'S TRY ONE

Multiply 2.6 × 10.

10 has _____ zero, so each digit moves _____ place to the left.

2.6 × 10 = _____

YOUR TURN — SKILL 6.1 · MULTIPLY

1 4.3 × 10

Product _____

2 0.8 × 10

Product _____

3 5.27 × 100

Product _____

4 0.6 × 100

Product _____

5 1.234 × 1000

Product _____

6 0.05 × 10

Product _____

7 0.09 × 1000

Product _____

8 A box holds **7.5 kg**. How many kg do **100** boxes hold?

Answer _____ kg

6.2 Decimal × Whole Number: Equal Groups

Goal: I can use equal groups and place value to multiply a decimal by a whole number.

SHOW HOW

Solve 2.4 × 3.

Group 1 (1) (1) (0.1) (0.1) (0.1) (0.1)

Group 2 (1) (1) (0.1) (0.1) (0.1) (0.1)

Group 3 (1) (1) (0.1) (0.1) (0.1) (0.1)

1. 2.4 × 3 means **3 groups of 2.4**. Each group is 2 ones and 4 tenths.
2. Count the ones: 3 × 2 = **6 ones**.
3. Count the tenths: 3 × 4 = **12 tenths**. 10 tenths make 1 one, so 12 tenths = **1.2**.
4. Put them together: 6 + 1.2 = **7.2**

Why it works: multiplying is adding equal groups: 2.4 + 2.4 + 2.4 = 7.2. Multiply each place, then regroup, just like in addition.

LET'S TRY ONE

Solve 1.3 × 4 with equal groups.

Ones: 4 × 1 = _____ ones Tenths: 4 × 3 = _____ tenths = _____

Product: _____ + _____ = _____

YOUR TURN — SKILL 6.2 · MULTIPLY EACH PLACE, THEN REGROUP

1 **0.3 × 2** (2 groups of 3 tenths)

Product _____

2 **0.4 × 5**

Product _____

3 **1.2 × 3**

Product _____

4 **2.3 × 2**

Product _____

5 **1.6 × 4**

Product _____

6 **2.7 × 3**

Product _____

7 **3.5 × 6**

Product _____

8 **1.5 × 4**

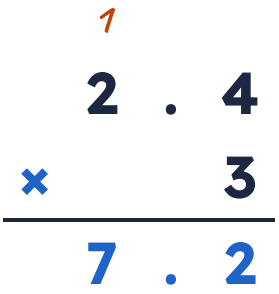
Product _____

6.3 Decimal × Whole Number: Written Method

Goal: I can multiply as whole numbers, then place the decimal point.

SHOW HOW

Solve 2.4 × 3 with the written method.



1. **Ignore the decimal point for now.** Multiply as whole numbers: 24 × 3 = **72**.
2. **Count decimal places** in the factors: 2.4 has **1**, 3 has **0**. Total: **1**.
3. Put **1** decimal place in the answer: 72 → **7.2**
4. **Check** with 6.2: 3 groups of 2.4 = 7.2 ✓. And 2.4 × 3 is about 2 × 3 = 6, so 7.2 makes sense.

Why it works: 2.4 is 24 tenths. 24 tenths × 3 = 72 tenths = 7.2. The answer is counted in tenths, so it has 1 decimal place.

LET'S TRY ONE

Solve 1.35 × 4.

As whole numbers: 135 × 4 = _____

Decimal places: _____ + _____ = _____ Product: _____

YOUR TURN — SKILL 6.3 · MULTIPLY AS WHOLE NUMBERS, THEN PLACE THE POINT

1 3.2 × 3

Product _____

2 4.5 × 2

Product _____

3 1.7 × 6

Product _____

4 0.26 × 3

Product _____

5 2.15 × 4

Product _____

6 6.08 × 5

Product _____

7 12.4 × 7

Product _____

8 0.125 × 8

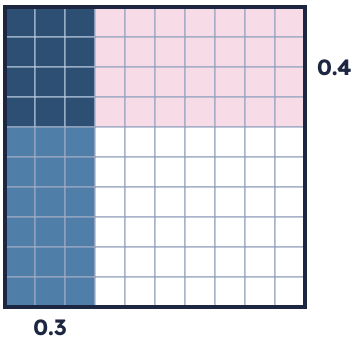
Product _____

6.4 Decimal × Decimal: Area Model

Goal: I can use a grid to see why tenths × tenths make hundredths.

SHOW HOW

Solve 0.3 × 0.4.

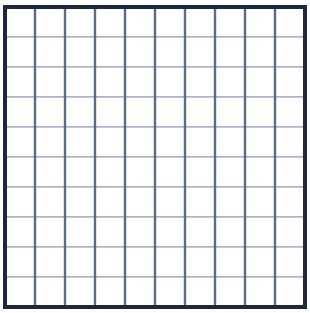


1. The grid is **1 whole**: 100 small squares.
2. Shade **0.3** of the width (3 columns, blue) and **0.4** of the height (4 rows, pink).
3. 0.3 × 0.4 is the part where they **overlap** (dark): 3 × 4 = **12 squares**.
4. Each square is 1 hundredth, so **0.3 × 0.4 = 0.12**

Why it works: "0.3 × 0.4" means "0.3 of 0.4". A tenth of a tenth is a hundredth, so tenths × tenths give hundredths. The product (0.12) is **smaller** than both factors, because you are taking part of a part.

LET'S TRY ONE

Shade the grid to solve 0.5 × 0.6: 5 columns and 6 rows.

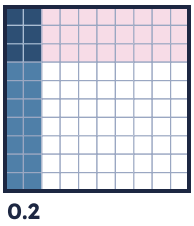


Overlap: _____ × _____ = _____ squares

Product: _____ hundredths = _____

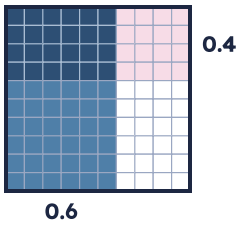
YOUR TURN — SKILL 6.4 · USE THE AREA MODEL

1 Write the product the model shows.



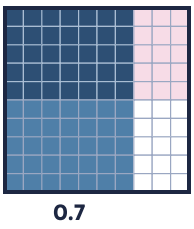
Product _____

2 Write the product the model shows.



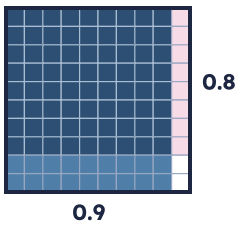
Product _____

3 Write the product the model shows.



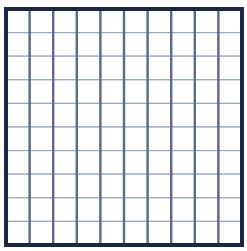
Product _____

4 Write the product the model shows.



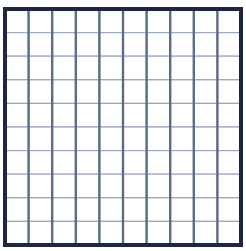
Product _____

5 Shade to solve **0.4 × 0.4**.



Product _____

6 Shade to solve **0.8 × 0.5**.



Product _____

7 **0.1 × 0.6** (sketch a grid if it helps)

Product _____

8 **0.3 × 0.9** (sketch a grid if it helps)

Product _____

6.5 Decimal × Decimal: Placing the Point

Goal: I can count decimal places to place the point in a product.

SHOW HOW

Solve 2.5 × 1.4.

1. **Multiply as whole numbers:** 25 × 14 = **350**. (25 × 10 = 250, 25 × 4 = 100, 250 + 100 = 350.)
2. **Count decimal places** in both factors: 2.5 has **1**, 1.4 has **1**. Total: **2**.
3. Put **2** decimal places in the answer: 350 → **3.50**, which is **3.5**. Too few digits? Write zeros in front: 0.2 × 0.3 → 6 → **0.06**.
4. **Check by estimating:** 2.5 × 1.4 is a bit more than 2.5 × 1 = 2.5, and less than 3 × 2 = 6. 3.5 fits.

Why it works: tenths × tenths = hundredths (6.4). 2.5 × 1.4 = 25 tenths × 14 tenths = 350 hundredths = 3.50. Count places **before** dropping trailing zeros.

LET'S TRY ONE

Solve 1.2 × 0.4.

As whole numbers: 12 × 4 = _____

Decimal places: _____ + _____ = _____ Product: _____

YOUR TURN — SKILL 6.5 · MULTIPLY, THEN COUNT DECIMAL PLACES

1 0.2 × 0.4

Product _____

2 1.5 × 0.3

Product _____

3 2.4 × 1.2

Product _____

4 0.6 × 0.05

Product _____

5 3.5 × 2.2

Product _____

6 0.12 × 0.4

Product _____

7 4.6 × 0.5

Product _____

8 1.25 × 1.6

Product _____

6.6 Estimate Products

Goal: I can round to estimate a product and catch a misplaced decimal point.

SHOW HOW

Estimate 4.8 × 3.1. Is 14.88 reasonable?

1. **Round each factor** to the nearest whole number: 4.8 ≈ **5**, 3.1 ≈ **3**.
2. **Multiply the rounded numbers:** 5 × 3 = **15**.
3. **Compare:** 14.88 is close to 15, so it is reasonable. An answer of 1.488 or 148.8 would mean the decimal point is in the wrong place.

Why it works: the rounded factors are close to the real ones, so the product is close too. Estimating is the fastest way to check where the decimal point goes.

LET'S TRY ONE

Estimate 6.2 × 2.9.

6.2 ≈ _____ 2.9 ≈ _____ Estimate: _____ × _____ = _____

Is 17.98 reasonable? _____

YOUR TURN — SKILL 6.6 · ESTIMATE BY ROUNDING TO WHOLE NUMBERS

1 **3.9 × 2**

Estimate _____

2 **5.2 × 4**

Estimate _____

3 **7.1 × 2.8**

Estimate _____

4 **9.6 × 3.3**

Estimate _____

5 **11.8 × 5.1**

Estimate _____

6 **2.45 × 6**

Estimate _____

7 Max says **4.2 × 4 = 168**. Estimate to decide if that is reasonable.

Reasonable? _____

8 Zoe says **3.2 × 1.95 = 6.24**. Estimate to decide if that is reasonable.

Reasonable? _____

6.7 Multiplication Word Problems

Goal: I can tell when a story needs multiplying, then solve and label.

SHOW HOW

One notebook costs $2.35. How much do 4 notebooks cost?

1. **What is asked?** The cost of 4 equal groups of $2.35: **multiply**. (“Each”, “per”, “equal groups”, “times as much”, and “of” for part of an amount: 0.5 of 3.2.)
2. **Estimate:** 2 × 4 = about **$8**.
3. **Multiply:** 235 × 4 = 940. Two decimal places: **9.40**.
4. **Check and label:** $9.40 is close to $8. Four notebooks cost **$9.40**.

Money tip: money keeps 2 decimal places, so write $9.40, not $9.4.

LET'S TRY ONE

A track is 1.8 km around. Ava runs 5 laps. How far does she run?

Operation: _____ Estimate: _____

Exact: _____ Answer with units: _____

YOUR TURN

SKILL 6.7 · SOLVE. LABEL YOUR ANSWER.

1 A juice box costs **$0.75**. How much do **3** juice boxes cost?

$ _____

2 A pot holds **2.5 L** of soup. How much soup is in **4** pots?

Answer _____ L

3 Leo walks **1.25 miles** to school each day. How far does he walk in **6** days?

Answer _____ mi

4 One bag of soil weighs **3.4 kg**. How much do **10** bags weigh?

Answer _____ kg

5 A ribbon is **4.5 m** long. Mia uses **0.5** of it. How many meters does she use?

Answer _____ m

6 A rug is **2.4 m** long and **1.5 m** wide. What is its area (length × width)?

Answer _____ sq m

7 Movie tickets cost **$3.85** each. How much do **4** tickets cost?

$ _____

8 A bag holds **0.6 kg** of nuts. Sam eats **0.5** of the bag. How many kg does he eat?

Answer _____ kg

Section 6 Review

Mixed practice from Skills 6.1 to 6.7. No hints: show what you know.

YOUR TURN **REVIEW · PROBLEMS 1–8**

1 **0.47 × 100**

Product _____

2 **3.6 × 2**

Product _____

3 **0.24 × 5**

Product _____

4 **0.7 × 0.6**

Product _____

5 **1.3 × 2.1**

Product _____

6 Estimate: **8.9 × 4.2**

Estimate _____

7 A pencil costs **$1.45**. How much do **3** pencils cost?

$ _____

8 A path is **0.8 km** long. A sign is **0.3** of the way along. How far along is the sign?

Answer _____ km

Parent Answer Key

Equal values are correct (3.50 = 3.5); money answers may keep two places ($9.40). “Try” is the Let's Try One problem.

6.1 Multiplying by 10, 100, and 1,000

Try **26** Multiplying by 10: each digit moves 1 place to the left (the point moves 1 right): 2.6 → 26.

1 **43** Multiplying by 10: each digit moves 1 place to the left (the point moves 1 right): 4.3 → 43.

2 **8** Multiplying by 10: each digit moves 1 place to the left (the point moves 1 right): 0.8 → 8.

3 **527** Multiplying by 100: each digit moves 2 places to the left (the point moves 2 right): 5.27 → 527.

4 **60** Multiplying by 100: each digit moves 2 places to the left (the point moves 2 right): 0.6 → 60.

5 **1234** Multiplying by 1000: each digit moves 3 places to the left (the point moves 3 right): 1.234 → 1234.

6 **0.5** Multiplying by 10: each digit moves 1 place to the left (the point moves 1 right): 0.05 → 0.5.

7 **90** Multiplying by 1000: each digit moves 3 places to the left (the point moves 3 right): 0.09 → 90.

8 **750 kg** Multiplying by 100: each digit moves 2 places to the left (the point moves 2 right): 7.5 → 750.

6.2 Decimal × Whole Number: Equal Groups

Try **5.2** 4 groups of 1.3: ones 4 × 1 = 4; tenths 4 × 3 = 12 tenths = 1.2. 4 + 1.2 = 5.2.

1 **0.6** 2 groups of 0.3: tenths 2 × 3 = 6 tenths = 0.6. Add: 0.6.

2 **2** 5 groups of 0.4: tenths 5 × 4 = 20 tenths = 2. Add: 2.

3 **3.6** 3 groups of 1.2: ones 3 × 1 = 3; tenths 3 × 2 = 6 tenths = 0.6. Add: 3.6.

4 **4.6** 2 groups of 2.3: ones 2 × 2 = 4; tenths 2 × 3 = 6 tenths = 0.6. Add: 4.6.

5 **6.4** 4 groups of 1.6: ones 4 × 1 = 4; tenths 4 × 6 = 24 tenths = 2.4. Add: 6.4.

6 **8.1** 3 groups of 2.7: ones 3 × 2 = 6; tenths 3 × 7 = 21 tenths = 2.1. Add: 8.1.

7 **21** 6 groups of 3.5: ones 6 × 3 = 18; tenths 6 × 5 = 30 tenths = 3. Add: 21.

8 **6** 4 groups of 1.5: ones 4 × 1 = 4; tenths 4 × 5 = 20 tenths = 2. Add: 6.

6.3 Decimal × Whole Number: Written Method

Try **5.4** Multiply as whole numbers: 135 × 4 = 540. Decimal places: 2 + 0 = 2, so 540 → 5.40 = 5.4.

1 **9.6** Multiply as whole numbers: 32 × 3 = 96. Decimal places: 1 + 0 = 1, so 96 → 9.6.

2 **9** Multiply as whole numbers: 45 × 2 = 90. Decimal places: 1 + 0 = 1, so 90 → 9.0 = 9.

3 **10.2** Multiply as whole numbers: 17 × 6 = 102. Decimal places: 1 + 0 = 1, so 102 → 10.2.

4 **0.78** Multiply as whole numbers: 26 × 3 = 78. Decimal places: 2 + 0 = 2, so 78 → 0.78.

5 **8.6** Multiply as whole numbers: 215 × 4 = 860. Decimal places: 2 + 0 = 2, so 860 → 8.60 = 8.6.

6 **30.4** Multiply as whole numbers: 608 × 5 = 3040. Decimal places: 2 + 0 = 2, so 3040 → 30.40 = 30.4.

7 **86.8** Multiply as whole numbers: 124 × 7 = 868. Decimal places: 1 + 0 = 1, so 868 → 86.8.

8 **1** Multiply as whole numbers: 125 × 8 = 1000. Decimal places: 3 + 0 = 3, so 1000 → 1.000 = 1.

6.4 Decimal × Decimal: Area Model

Try **0.3** 5 columns × 6 rows overlap in 30 squares = 30 hundredths = 0.30 = 0.3.

1 **0.06** 2 columns × 3 rows overlap in 6 squares = 6 hundredths = 0.06.

2 **0.24** 6 columns × 4 rows overlap in 24 squares = 24 hundredths = 0.24.

3 **0.35** 7 columns × 5 rows overlap in 35 squares = 35 hundredths = 0.35.

4 **0.72** 9 columns × 8 rows overlap in 72 squares = 72 hundredths = 0.72.

5 **0.16** 4 columns × 4 rows overlap in 16 squares = 16 hundredths = 0.16.

6 **0.4** 8 columns × 5 rows overlap in 40 squares = 40 hundredths = 0.4.

7 **0.06** 1 columns × 6 rows overlap in 6 squares = 6 hundredths = 0.06.

8 **0.27** 3 columns × 9 rows overlap in 27 squares = 27 hundredths = 0.27.

6.5 Decimal × Decimal: Placing the Point

Try **0.48** Multiply as whole numbers: 12 × 4 = 48. Decimal places: 1 + 1 = 2, so 48 → 0.48.

1 **0.08** Multiply as whole numbers: 2 × 4 = 8. Decimal places: 1 + 1 = 2, so 8 → 0.08.

2 **0.45** Multiply as whole numbers: 15 × 3 = 45. Decimal places: 1 + 1 = 2, so 45 → 0.45.

3 **2.88** Multiply as whole numbers: 24 × 12 = 288. Decimal places: 1 + 1 = 2, so 288 → 2.88.

4 **0.03** Multiply as whole numbers: 6 × 5 = 30. Decimal places: 1 + 2 = 3, so 30 → 0.030 = 0.03.

5 **7.7** Multiply as whole numbers: 35 × 22 = 770. Decimal places: 1 + 1 = 2, so 770 → 7.70 = 7.7.

6 **0.048** Multiply as whole numbers: 12 × 4 = 48. Decimal places: 2 + 1 = 3, so 48 → 0.048.

7 **2.3** Multiply as whole numbers: 46 × 5 = 230. Decimal places: 1 + 1 = 2, so 230 → 2.30 = 2.3.

8 **2** Multiply as whole numbers: 125 × 16 = 2000. Decimal places: 2 + 1 = 3, so 2000 → 2.000 = 2.

6.6 Estimate Products

Try **18** Round each number to the nearest whole: 6.2 ≈ 6, 2.9 ≈ 3. 6 × 3 = 18.

1 **8** Round each number to the nearest whole: 3.9 ≈ 4, 2 ≈ 2. 4 × 2 = 8.

2 **20** Round each number to the nearest whole: 5.2 ≈ 5, 4 ≈ 4. 5 × 4 = 20.

3 **21** Round each number to the nearest whole: 7.1 ≈ 7, 2.8 ≈ 3. 7 × 3 = 21.

4 **30** Round each number to the nearest whole: 9.6 ≈ 10, 3.3 ≈ 3. 10 × 3 = 30.

5 **60** Round each number to the nearest whole: 11.8 ≈ 12, 5.1 ≈ 5. 12 × 5 = 60.

6 **12** Round each number to the nearest whole: 2.45 ≈ 2, 6 ≈ 6. 2 × 6 = 12.

7 **No (it should be 16.8)** Estimate: 4 × 4 = 16. 168 is far from 16. Exact: 4.2 × 4 = 16.8.

8 **Yes** Estimate: 3 × 2 = 6. 6.24 is close to 6. Exact: 3.2 × 1.95 = 6.24.

6.7 Multiplication Word Problems

Try **9** 5 laps of 1.8 km: multiply. 18 × 5 = 90, 1 decimal place → 9.0 = 9 km.

1. **$2.25** Multiply as whole numbers: 75 × 3 = 225. Decimal places: 2 + 0 = 2, so 225 → 2.25.
2. **10 L** Multiply as whole numbers: 25 × 4 = 100. Decimal places: 1 + 0 = 1, so 100 → 10.0 = 10.
3. **7.5 mi** Multiply as whole numbers: 125 × 6 = 750. Decimal places: 2 + 0 = 2, so 750 → 7.50 = 7.5.
4. **34 kg** Multiply as whole numbers: 34 × 10 = 340. Decimal places: 1 + 0 = 1, so 340 → 34.0 = 34.
5. **2.25 m** Multiply as whole numbers: 45 × 5 = 225. Decimal places: 1 + 1 = 2, so 225 → 2.25.
6. **3.6 sq m** Multiply as whole numbers: 24 × 15 = 360. Decimal places: 1 + 1 = 2, so 360 → 3.60 = 3.6.
7. **$15.40** Multiply as whole numbers: 385 × 4 = 1540. Decimal places: 2 + 0 = 2, so 1540 → 15.40.
8. **0.3 kg** Multiply as whole numbers: 6 × 5 = 30. Decimal places: 1 + 1 = 2, so 30 → 0.30 = 0.3.

Review Section 6 Review

1. **47** Multiplying by 100: each digit moves 2 places to the left (the point moves 2 right): 0.47 → 47.
2. **7.2** Multiply as whole numbers: 36 × 2 = 72. Decimal places: 1 + 0 = 1, so 72 → 7.2.
3. **1.2** Multiply as whole numbers: 24 × 5 = 120. Decimal places: 2 + 0 = 2, so 120 → 1.20 = 1.2.
4. **0.42** Multiply as whole numbers: 7 × 6 = 42. Decimal places: 1 + 1 = 2, so 42 → 0.42.
5. **2.73** Multiply as whole numbers: 13 × 21 = 273. Decimal places: 1 + 1 = 2, so 273 → 2.73.
6. **36** Round each number to the nearest whole: 8.9 ≈ 9, 4.2 ≈ 4. 9 × 4 = 36.
7. **$4.35** Multiply as whole numbers: 145 × 3 = 435. Decimal places: 2 + 0 = 2, so 435 → 4.35.
8. **0.24 km** Multiply as whole numbers: 8 × 3 = 24. Decimal places: 1 + 1 = 2, so 24 → 0.24.